13 The number of animals in a population at the start of year t is P_t
The number of animals at the start of year 1 is 400

Given that

$$P_{t+1} = 1.01P_t$$

work out the number of animals at the start of year 3

.....

(Total for Question 13 is 2 marks)

14 y is inversely proportional to x^3

$y = 44$ when $x = a$

Show that $y = 5.5$ when $x = 2a$

$y = \dfrac{k}{x^3}$

1 mark
Write the inverse-cube equation.

When $x = a$: $44 = \dfrac{k}{a^3}$

1 mark
Substitute the given pair.

so $k = 44a^3$.

When $x = 2a$: $y = \dfrac{44a^3}{(2a)^3}$

$y = \dfrac{44a^3}{8a^3} = 44 \div 8 = 5.5$

1 mark
Show the substitution, cancellation and stated result.

(Total for Question 14 is 3 marks)

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